Sardar Vallabhbhai National Institute of Technology (SVNIT), Surat
Department of Artificial Intelligence
Intelligent Multi-Agent and Expert Systems (AI401), Semester VII

Instructions

1. Save your file as LAB_No_Roll_No.doc (e.g. LAB_11_U21AI001.doc).
2. Paste a snapshot of each step followed along with the result(s).
3. Record your observations and comments after the results.

Experiment 11

Expert System Application — Medical Diagnosis with Certainty Factors

Aim

To build a diagnostic expert system whose rules carry certainty factors, produce a ranked differential diagnosis for a patient whose symptoms are themselves reported with uncertainty, and drive the consultation so that only the questions needed to reach a conclusion are asked.

Theory

Medical rules are rarely certain in either direction: a symptom supports a diagnosis without implying it, and a patient reports a symptom with their own degree of confidence. A certainty factor attaches a number between −1 and +1 to each, positive for evidence in favour and negative for evidence against, with zero meaning nothing is known. Three combination rules are then enough to run the system. The strength of a conjunction is the minimum of its parts, so a chain is as weak as its weakest link; the contribution of a rule is its own certainty factor multiplied by that strength, so a strong rule on weak evidence yields a weak conclusion; and two rules supporting the same hypothesis combine as CF1 + CF2 – CF1·CF2, which accumulates towards one without ever reaching it. The combination is commutative and associative, so the order in which evidence arrives cannot change the result.

The output is a ranked differential rather than a single answer, and a threshold decides when the system is confident enough to conclude. That design has a second consequence worth measuring. Since each rule names only a few findings, the system need not ask every question on the form: it can pursue the hypothesis that currently leads and ask only what would confirm it, stopping as soon as the threshold is passed. The number of questions a consultation costs then depends on the order in which they are asked, which is a property of the control strategy and not of the knowledge base.

Algorithm

Algorithm 1 — Rule base and certainty-factor propagation

1. Write a rule base of at least eight rules over four candidate diagnoses, each rule a conjunction of findings — some required present and some required absent — together with its own certainty factor. Represent a patient as a certainty factor per finding, positive when the patient reports it and negative when the patient denies it.
2. Implement the three combination functions: the minimum for a conjunction, the product of the rule certainty and the antecedent strength for a single rule, and the parallel combination for two rules supporting the same diagnosis. For a given patient, evaluate every rule, combine by diagnosis and print the ranked differential with the rules that contributed to each entry.

Algorithm 2 — Driving the consultation

1. Implement a consultation that starts with nothing known and asks one finding at a time, re-evaluating the differential after each answer and stopping when the leading diagnosis passes a stated threshold or no question remains. Record the questions asked in order.
2. Provide three question-selection strategies — the fixed order of an intake form, the order of how often a finding appears in the rule base, and a goal-directed order that asks only findings appearing in rules for the currently leading diagnosis — and report for one patient the questions each strategy asks and the conclusion each reaches.

Procedure and Result

Paste the snapshots of the steps followed and the output produced by the program below, including the rule base, the patient findings, the ranked differential with the contributing rules and the combined certainty factors, and the question sequence and conclusion of each of the three strategies for the same patient.

Observations

Comment on how the combined certainty factor of the leading diagnosis compares with the certainty factor of either rule alone, on what a conjunction evaluated as a minimum does to a diagnosis resting on one weakly reported finding, on why a patient who denies a symptom is not the same as a patient about whom nothing is known, and on what the system should report when no diagnosis passes the threshold.

Exercises

1. Generate a population of patients and perturb every rule certainty factor by a random amount within plus or minus 0.05, 0.10 and 0.20, as an expert giving slightly different numbers on a different day would. Report at each level how often the leading diagnosis changes, the mean change in its certainty factor, and how often a consultation that previously passed the threshold no longer does.
2. Run all three question strategies over the whole population and report the mean and maximum number of questions each needs and whether any of them reaches a different conclusion. State which strategy is best for the patient demonstrated in Algorithm 2, whether that strategy is also best on average, and what that implies about choosing a question order for a real clinic.